SINYAL DAN SISTEM (2NK)

No 2022
Date 01 November

SISTEM LINIER

memenuhi syarat : a). Homogenitas
b). Additive

> Sifat Homogenitas :

$x[n] \rightarrow [SL] \rightarrow y[n]$

$Ax[n] \rightarrow [SL] \rightarrow Ay[n]$

misal $y(n) = 5x(n)$

misal $x(n) = \sin(n)$

$\therefore y(n) = 5x(n) = 5.\sin(n)$

$x'(n) = Ax(n) = A.\sin(n)$

$y'(n) = 5x'(n) = 5A\sin(n)$

$= A5\sin(n) = A\,y(n)$

($5\sin(n) = y(n)$)

↓ Sistem homogen

* A = Konstanta

> Sifat Additive :

$x_1(n) \rightarrow [SL] \rightarrow y_1(n)$

$x_2(n) \rightarrow [SL] \rightarrow y_2(n)$

$x_1(n) + x_2(n) \rightarrow [SL] \rightarrow y_1(n) + y_2(n)$

misal $x_1(n) = \sin(n)$, $x_2(n) = \cos(n)$

$y_1(n) = 5x_1(n) = 5\sin(n)$

$y_2(n) = 5x_2(n) = 5\cos(n)$

misal $x'(n) = x_1(n) + x_2(n)$

$= \sin(n) + \cos(n)$

$\therefore y'(n) = 5x'(n) = 5[\sin(n) + \cos(n)]$

$= 5\sin(n) + 5\cos(n)$

$= y_1(n) + y_2(n)$

↓ Sistem Additive

Karena $y(n) = 5x(n)$ terbukti homogen dan additive, maka $y(n) = 5x(n)$ bersifat sistem linier.

Contoh : Tentukan apakah $y(n) = 5x(n)+2$ bersifat linier

a). Cek homogenitas

misal $x(n) = \sin(n)$

$y(n) = 5\sin(n) + 2$

misal $x'(n) = Ax(n) = A\sin(n)$

$\therefore y'(n) = 5x'(n) + 2$

$= 5A\sin(n) + 2$

$= A\,y(n) + 2$

$\neq A\,y(n)$

maka, $y(n) = 5x(n) + 2$ tidak homogen.

b). Cek Additive

misal $x_1(n) = \sin(n)$, $x_2 = \cos(n)$

$y_1(n) = 5x_1(n) + 2 = 5\sin(n) + 2$

$y_2(n) = 5x_2(n) + 2 = 5\cos(n) + 2$

$\therefore x'(n) = x_1(n) + x_2(n)$

$= \sin(n) + \cos(n)$

$y'(n) = 5x'(n) + 2$

$= 5[\sin(n) + \cos(n)] + 2$

$= 5\sin(n) + 5\cos(n) + 2$

$= [5\sin(n) + 2] + [5\cos(n) + 2] - 2$

($5\sin(n) + 2 = y_1(n)$, $5\cos(n) + 2 = y_2(n)$)

$\neq y_1(n) + y_2(n)$

maka, $y(n) = 5x(n) + 2$ tidak additive.

$y(n) = 5x(n) + 2$ tidak linier

Latihan : Tentukan apakah $y(n) = 1/x(n)$ bersifat linier

a). Cek homogenitas

misal $x(n) = \sin(n)$

$y(n) = \frac{1}{\sin(n)}$

misal $x'(n) = Ax(n) = A\sin(n)$

$\therefore y'(n) = \frac{1}{x'(n)} = \frac{1}{A\sin(n)}$

$= \frac{1}{A} \cdot \frac{1}{\sin(n)}\, y(n)$

$= \frac{y(n)}{A} \neq A\,y(n)$

maka, $y(n) = 1/x(n)$ tidak homogen

b). Cek Additive

misal $x_1(n) = \sin(n)$, $x_2(n) = \cos(n)$

$y_1(n) = \frac{1}{x_1(n)} = \frac{1}{\sin(n)}$

$y_2(n) = \frac{1}{x_2(n)} = \frac{1}{\cos(n)}$

$\therefore x'(n) = x_1(n) + x_2(n)$

$= \sin(n) + \cos(n)$

$y'(n) = \frac{1}{x'(n)}$

$= \frac{1}{\sin(n) + \cos(n)}$

$\neq \frac{1}{\sin(n)} + \frac{1}{\cos(n)}$

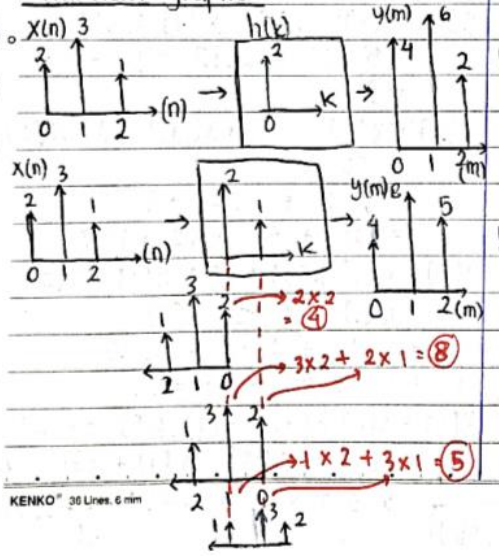
$\neq y_1(n) + y_2(n)$

maka, $y(n) = 1/x(n)$ tidak additive.

$y(n) = 1/x(n)$ tidak linier

KONVOLUSI 1D PADA SINYAL DISKRIT

Ilustrasi grafik

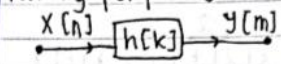


Notasinya : $y[m] = x[n] * h[k]$

output, input, impulse respone

Secara matematis : $y[m] = \sum_{n=0}^{N-1} x[n].h[m-n]$

Filtering perspective :



contoh soal :

Diket $x[n] = [4, 3, 2]$ ($N = 3$)

$h[k] = [5, 2, 1]$ ($k = 3$)

$M = N + k - 1 = 3 + 3 - 1 = 5$

Hitung nilai y[m] !

$y[0] = x[n].h[m-n] = x[0].h[0-0]$

$= 4.5 = 20$

$y[1] = x[0].h[1-0] + x[1].h[1-1]$

$= x[0].h[1] + x[1].h[0]$

$= 4.2 + 3.5 = 23$

$y[2] = x[0].h[2-0] + x[1].h[2-1] + x[2].h[2-2]$

$= x[0].h[2] + x[1].h[1] + x[2].h[0]$

$= 4.1 + 3.2 + 2.5 = 20$

$y[3] = x[0].h[3-0] + x[1].h[3-1] + x[2].h[3-2] + x[3].h[3-3]$

$= x[0].\cancel{h[3]} + x[1].h[2] + x[2].h[1] + \cancel{x[3]}.h[0]$

$= 3.1 + 2.2 = 7$

$y[4] = x[0].h[4-0] + x[1].h[4-1] + x[2].h[4-2] + x[3].h[4-3] + x[4].h[4-4]$

$= x[0].\cancel{h[4]} + x[1].\cancel{h[3]} + x[2].h[2] + \cancel{x[3]}.h[1] + \cancel{x[4]}.h[0]$

$= 2.1 = 2$

$y[m] = [20, 23, 20, 7, 2]$

$y[m] = \sum_{k=0}^{k=1} h[k] \cdot x[m-k]$

$y[0] = h[0] \cdot x[0-0] = 5.4 = 20$

$y[1] = h[0].x[1] + h[1].x[0]$
$= 5.3 + 2.4 = 23$

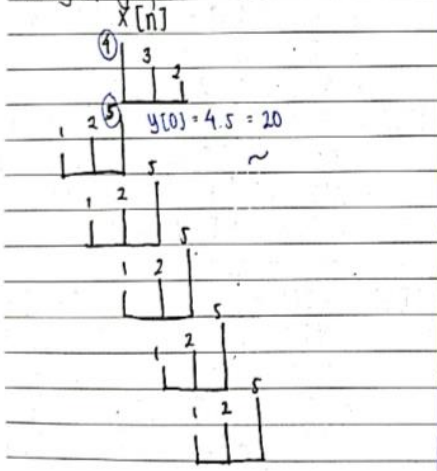
$y[2] = h[0].x[2] + h[1].x[1] + h[2].x[0]$
$= 5.2 + 2.3 + 1.4$
$= 20$

$y[3] = h[0].\cancel{x[3]} + h[1].x[2] + h[2] x[1] + \cancel{h[3]}.x[0]$
$= 2.2 + 1.3 = 7$

$y[4] = h[0].\cancel{x[4]} + h[1].\cancel{x[3]} + h[2].x[2] + \cancel{h[3]}x[1] + \cancel{h[4]}.x[0]$
$= 2.1 = 2$

Hasilnya sama, yaitu
$y[m] = [20, 23, 20, 7, 2]$

Dengan grafik:



Diket $x[n] = [5, 2, 1]$
$h_1[k] = [3, 2, 1]$, $h_2[j] = [4, 4, 3]$

Hitung nilai $y[m] = x[n] * (h_1[k] + h_2[j])$

jawab:

$(h_1[k] \cdot h_2[j])$

$k=0 \rightarrow h_1[0] \cdot h_2[0]$
$= 3 \cdot 4 = 12$

$k=1 \rightarrow h_1[0].h_2[1] + h_1[1].h_2[0]$
$= 3 \cdot 4 + 2.4 = 20$

$k=2 \rightarrow h_1[0].h_2[2] + h_1[1].h_2[1] + h_1[2].h_2[0]$
$= 3.3 + 2.4 + 1.4 = 21$

$k=3 \rightarrow h_1[0].\cancel{h_2[3]} + h_1[1].h_2[2] + h_1[2].h_2[1] + \cancel{h_1[3]}.h_2[0]$
$= 2.3 + 1.4 = 10$

$k=4 \rightarrow h_1[0].\cancel{h_2[4]} + h_1[1].\cancel{h_2[3]} + h_1[2].h_2[2] + \cancel{h_1[3]}.h_2[1] + \cancel{h_1[4]}.h_2[0]$
$= 1.3 = 3$

$h_1[k] \cdot h_2[j] = [12, 20, 21, 10, 3]$
(0 1 2 3 4)

$y[m] = x[n].(h_1[k].h_2[j])$

$y[0] = x[0] \cdot h[0] = 5.12 = 60$

$y[1] = x[0].h[1] + x[1].h[0]$
$= 5.20 + 2.12 = 100 + 24$
$= 124$

$y[2] = x[0].h[2] + x[1].h[1] + x[2].h[0]$
$= 5.21 + 2.20 + 1.12$
$= 105 + 40 + 12 = 157$

$y[3] = x[0].h[3] + x[1].h[2] + x[2].h[1] + \cancel{x[3]}.h[0]$
$= 5.10 + 2.21 + 1.20$
$= 50 + 42 + 20 = 112$

$y[4] = x[0].h[4] + x[1].h[3] + x[2].h[2] + \cancel{x[3]}.h[1] + \cancel{x[4]}.h[0]$
$= 5.3 + 2.10 + 1.21 = 15 + 20 + 21$
$= 56$

$y[m] = [60, 84, 157, 112, 56]$

KENKO® 36 Lines, 6 mm

SINYAL DAN SISTEM

No Selasa
Date 15 November 2022

KORELASI

$y[m] = x[n] \oplus t[k]$

Secara matematis:

$$y[m] = \sum_{m=0}^{M-1} \sum_{k=0}^{K-1} x[(m) - (K-1)] \cdot t[k]$$

Diketahui $x[n] = [5, 4, 2]$
$t[k] = [3, 2, 1]$

Hitung nilai korelasinya!

jawab:

$y[0] = x[0 - (3-1)] \cdot t[0] + x[1 - (3-1)] \cdot t[1] + x[2 - (3-1)] \cdot t[2]$
$= x[\cancel{-2}] \cdot t[0] + x[\cancel{-1}]\, t[1] + x[0]\, t[2]$
$= 5 \cdot 1 = 5$

$y[1] = x[1 - 2] \cdot t[0] + x[2 - 2] \cdot t[1] + x[3 - 2] t[2]$
$= x[\cancel{-1}]\, t[0] + x[0] \cdot t[1] + x[1] \cdot t[2]$
$= 5.2 + 4.1 = 10 + 4 = 14$

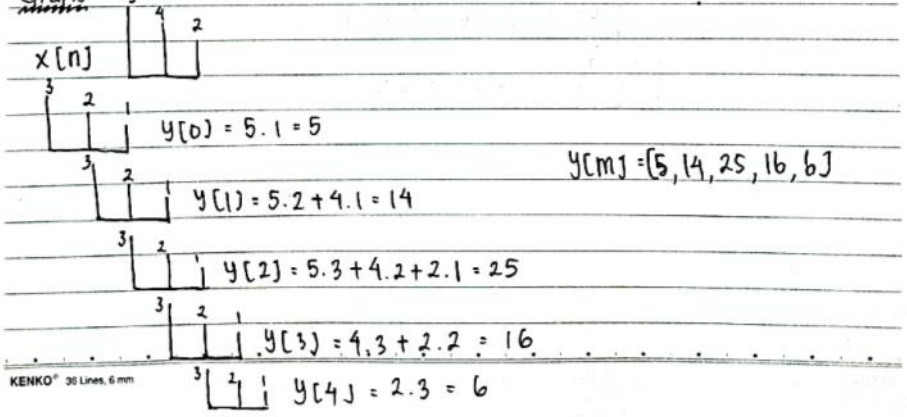
$y[2] = x[2 - 2]\, t[0] + x[3 - 2]\, t[1] + x[4 - 2]\, t[2]$
$= x[0]\, t[0] + x[1]\, t[1] + x[2]\, t[2]$
$= 5.3 + 4.2 + 2.1 = 15 + 8 + 2 = 25$

$y[3] = x[3 - 2]\, t[0] + x[4 - 2]\, t[1] + x[5 - 2]\, t[2]$
$= x[1]\, t[0] + x[2]\, t[1] + x[\cancel{3}]\, t[2]$
$= 4.3 + 2.2 = 12 + 4 = 16$

$y[4] = x[4 - 2]\, t[0] + x[5 - 2] \cdot t[1] + x[6 - 2] \cdot t[2]$
$= x[2] \cdot t[0] + x[\cancel{3}] \cdot t[1] + x[\cancel{4}] \cdot t[2]$
$= 2 \cdot 3 = 6$

$y[m] = [5, 14, 25, 16, 6]$

Grafis



KENKO® 36 Lines, 6 mm

$$y[m] = \sum_{m=0,n=0}^{M-1\ N-1} t[m-(N-i)]\ x[n]$$

$y[0] = t[0-(3-1)] \times [0] + t[1-(3-1)] \times [1] + t[2-(3-1)] \times [2]$
$= t[-2] \times [0] + t[-1] \times [1] + t[0] \times [2] = 3.2 = 6$

$y[1] = t[1-2] \times [0] + t[2-2] \times [1] + t[3-2] \times [2]$
$= t[-1] \times [0] + t[0] \times [1] + t[1] \times [2] = 3.4 + 2.2$
$= 16$

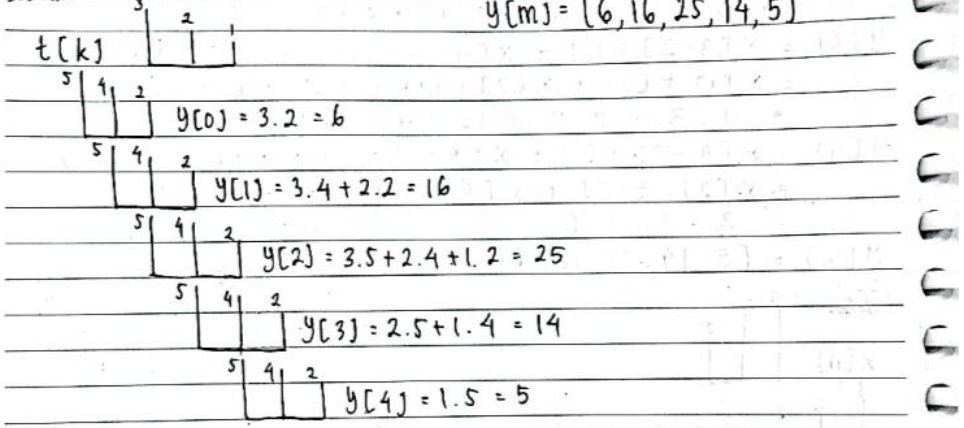
$y[2] = t[2-2] \times [0] + t[3-2] \times [1] + t[4-2] \times [2]$
$= t[0] \times [0] + t[1] \times [1] + t[2] \times [2]$
$= 3.5 + 2.4 + 1.2 = 25$

$y[3] = t[3-2] \times [0] + t[4-2] \times [1] + t[5-2] \times [2]$
$= t[1] \times [0] + t[2] \times [1] + t[3] \times [2]$
$= 2.5 + 1.4 = 14$

$y[4] = t[4-2] \times [0] + t[5-2] \times [1] + t[6-2] \times [2]$
$= t[2] \times [0] + t[3] \times [1] + t[4] \times [2]$
$= 1.5 = 5$

Grafis

$y[m] = [6, 16, 25, 14, 5]$



KONVOLUSI 2D

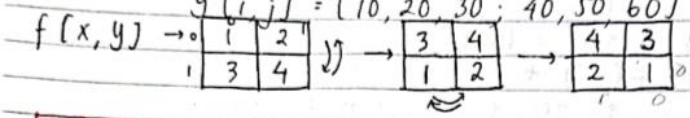
Notasinya : $y[m,n] = g(i,j) * f[x,y]$

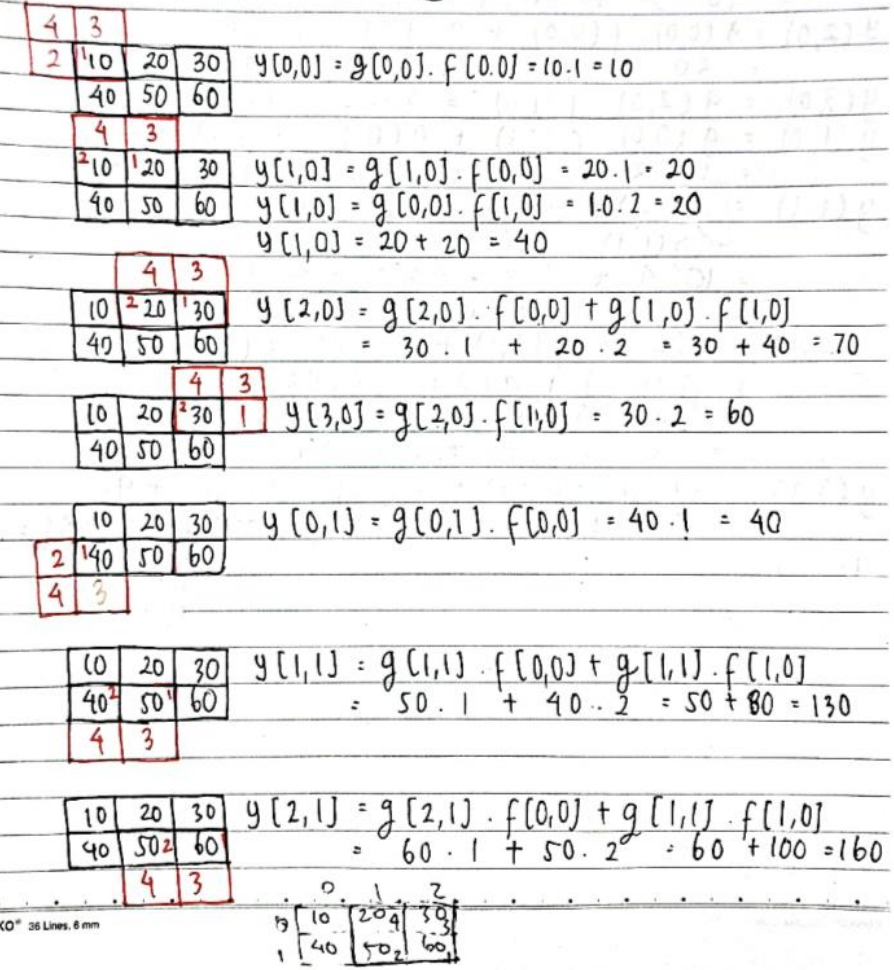
$g(i,j)$: sbg koordinat; $f[x,y]$: yg bergerak

Secara grafis

Diketahui $f[x,y] = [1,2 ; 3,4]$

$g[i,j] = [10, 20, 30 ; 40, 50, 60]$





10	20	30	
40	50	60	1
		4	3

$y[2,1] = g[2,1] \cdot f[1,0]$
$= 60 \cdot 2 = 120$

	0	1	2	3
0	10	40	70	60
1	70	230	330	240
2	120	310	380	240

$y(0,0) = g(0,0) \cdot f(0,0) = 10 \cdot 1 = 10$ //

$y(1,0) = g(0,0) \cdot f(1,0) + g(1,0) \cdot f(0,0)$
$= 10 \cdot 2 + 20 \cdot 1 = 20 + 20 = 40$ //

$y(2,0) = g(2,0) \cdot f(0,0) + g(1,0) \cdot f(1,0)$
$= 30 \cdot 1 + 20 \cdot 2 = 30 + 40 = 70$ //

$y(3,0) = g(2,0) \cdot f(1,0) = 30 \cdot 2 = 60$ //

$y(0,1) = g(0,0) \cdot f(0,1) + g(0,1) \cdot f(0,0)$
$= 10 \cdot 3 + 40 \cdot 1 = 30 + 40 = 70$ //

$y(1,1) = g(0,0) \cdot f(1,1) + g(1,0) \cdot f(0,1) + g(0,1) \cdot f(1,0) + g(1,1) \cdot f(0,0)$
$= 10 \cdot 4 + 20 \cdot 3 + 40 \cdot 2 + 50 \cdot 1$
$= 40 + 60 + 80 + 50 = 230$ //

$y(2,1) = g(1,0) \cdot f(1,1) + g(2,0) \cdot f(0,1) + g(1,1) \cdot f(1,0) + g(2,1) \cdot f(0,0)$
$= 20 \cdot 4 + 30 \cdot 3 + 50 \cdot 2 + 60 \cdot 1$
$= 80 + 90 + 100 + 60 = 330$ //

$y(3,1) = g(2,0) \cdot f(1,1) + g(2,1) \cdot f(1,0)$
$= 30 \cdot 4 + 60 \cdot 2 = 120 + 120 = 240$ //

$y(0,2) =$

SINYAL DAN SISTEM

No Soal 4
Date 22 NOV 2022

FOURIER TRANSFORM

Fourier → transformasi dari domain waktu ke domain fourier dan begitu juga sebaliknya.

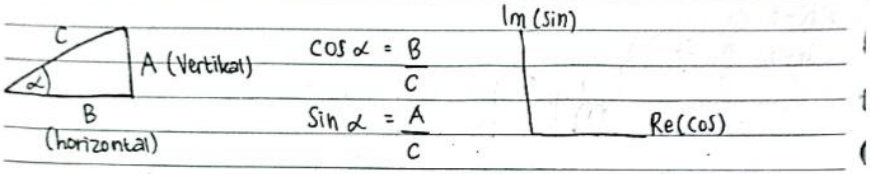
Waktu (t) ⇄ (FT / FT) Frekuensi (f)

$$Re\,X[k] = \sum_{i=0}^{N-1} X[i] \cos(2\pi ki/N)$$

$$Im\,X[k] = -\sum_{i=0}^{N-1} X[i] \sin(2\pi ki/N)$$

} mengadopsi dari bilangan kompleks

k = dari 0 sd N/2



$\cos\alpha = \frac{B}{C}$

$\sin\alpha = \frac{A}{C}$

Contoh: $X[n] = [1, 2, 3, 4]$ $N = 4$
Cari nilai $Re\,X[k]$ dan $Im\,X[k]$?

Jawab: $k = 0 - N/2 \Rightarrow 0 - 4/2 \Rightarrow 0 - 2$

$Re\,X[0] = X[0]\cos(2\pi.0.0/4) + X[1]\cos(2\pi.0.1/4) + X[2]\cos(2\pi.0.2/4) + X[3]\cos(2\pi.0.3/4)$
$= 1\cos 0 + 2\cos 0 + 3\cos 0 + 4\cos 0$
$= 1.1 + 2.1 + 3.1 + 4.1 = 10$ //

$Re\,X[1] = X[0]\cos(2\pi.1.0/4) + X[1]\cos(2\pi.1.1/4) + X[2]\cos(2\pi.1.2/4) + X[3]\cos(2\pi.1.3/4)$
$= 1\cos 0 + 2\cos\frac{\pi}{2} + 3\cos\pi + 4\cos\frac{3\pi}{2}$
$= 1.1 + 2.0 + 3(-1) + 4.0$
$= 1 - 3 = -2$ //

$Re\,X[2] = X[0]\cos(2\pi.2.0/4) + X[1]\cos(2\pi.2.1/4) + X[2]\cos(2\pi.2.2/4) + X[3]\cos(2\pi.2.3/4)$
$= 1\cos 0 + 2\cos\pi + 3\cos 2\pi + 4\cos 3\pi$
$= 1.1 + 2(-1) + 3.1 + 4(-1)$
$= -2$ //

$Im\ X[0] = -[X[0]\sin(2\pi.0.0/4) + X[1]\sin(2\pi.0.1/4) + X[2]\sin(2\pi.0.2/4) + X[3]\sin(2\pi.0.3/4)]$
$= -[1\sin 0 + 2\sin 0 + 3\sin 0 + 4\sin 0]$
$= -[1.0 + 2.0 + 3.0 + 4.0] = 0$ //

$Im\ X[1] = -[X[0]\sin(2\pi.1.0/4) + X[1]\sin(2\pi.1.1/4) + X[2]\sin(2\pi.1.2/4) + X[3]\sin(2\pi.1.3/4)]$
$= -[1\sin 0 + 2\sin\frac{\pi}{2} + 3\sin\pi + 4\sin\frac{3\pi}{2}]$
$= -[1.0 + 2.1 + 3.0 + 4.(-1)]$
$= -(-2) = 2$ //

$Im\ X[2] = -[X[0]\sin(2\pi.2.0/4) + X[1]\sin(2\pi.2.1/4) + X[2]\sin(2\pi.2.2/4) + X[3]\sin(2\pi.2.3/4)]$
$= -[1\sin 0 + 2\sin\pi + 3\sin 2\pi + 4\sin 3\pi]$
$= -[1.0 + 2.0 + 3.0 + 4.0]$
$= 0$ //

INVERS FOURIER TRANSFORM

$Re\ \tilde{X}[k] = \frac{Re\ X[k]}{N/2}$ $\qquad Im\ \tilde{X}[k] = -\frac{Im\ X[k]}{N/2}$

Kecuali khusus untuk:
$Re\ \tilde{X}[0] = \frac{Re\ X[k]}{N}$ / $Re\ \tilde{X}[N/2] = \frac{Re\ X[N/2]}{N/2}$
$Im\ \tilde{X}[0] = -\frac{Im\ X[k]}{N}$

$IFT \rightarrow x[i] = \sum_{k=0}^{N/2} Re\ \tilde{X}[k]\cos(2\pi.k.i/N) + \sum_{k=0}^{N/2} Im\ \tilde{X}[k]\sin(2\pi.k.i/N)$

1. $Mag(k) = \sqrt{Re(k)^2 + Im(k)^2}$
2. $Re(\ ) = [Re(0) \ldots Re(k)]$
3. $Im(\ ) = [Im(0) \ldots Im(k)]$
4. $\angle\theta\ phoe(k) = \tan^{-1}\left[\frac{Im(k)}{Re(k)}\right]$

} Forward Fourier Transform

$x[n] = \{1,2,3,4\}$ > konversi $x \rightarrow \tilde{x}$ $\qquad 0\ \ 1\ \ 2 \rightarrow N/2$
$Re\ X(k) = \{10, -2, -2\}$ $\qquad Re\ \tilde{X}(k) = \{2{,}5;\ -1;\ -0{,}5\}$
$Im\ X(k) = \{0, 2, 0\}$ $\qquad Im\ \tilde{X}(k) = (0, -1, 0\}$

$X(0) = Re\ \tilde{X}(0)\cos(2\pi.0.0/4) + Re\ \tilde{X}(1)\cos(2\pi.1.0/4) + Re\ \tilde{X}(2)\cos(2\pi.2.0/4) + Im\ \tilde{X}(0)\sin(2\pi.0.0/4) + Im\ \tilde{X}(1)\sin(2\pi.1.0/4) + Im\ \tilde{X}(2)\sin(2\pi.2.0/4)$.
$= (2{,}5)(1) + (-1)(1) + (-0{,}5)(1) + 0 + (-1).0 + 0.0 = 2{,}5 - 1{,}5 = 1$ //

$X(1) = Re\ \tilde{X}(0)\cos(2\pi.0.1/4) + Re\ \tilde{X}(1)\cos(2\pi.1.1/4) + Re\ \tilde{X}(2)\cos(2\pi.2.1/4) + Im\ \tilde{X}(0)\sin(2\pi.0.1/4) + Im\ \tilde{X}(1)\sin(2\pi.1.1/4) + Im\ \tilde{X}(2)\sin(2\pi.2.1/4)$
$= (2{,}5)(1) + (-1)(0) + (-0{,}5)(-1) + 0.0 + (-1)(1) + 0.0$
$= 2{,}5 + 0{,}5 - 1 = 2$ //

$X(2) = Re\ \tilde{X}(0)\cos(2\pi.0.2/4) + Re\ \tilde{X}(1)\cos(2\pi.1.2/4) + Re\ \tilde{X}(2)\cos(2\pi.2.2/4) + Im\ \tilde{X}(0)\sin(2\pi.0.2/4) + Im\ \tilde{X}(1)\sin(2\pi.1.2/4) + Im\ \tilde{X}(2)\sin(2\pi.2.2/4)$
$= (2{,}5)(1) + (-1)(-1) + (-0{,}5)(1) + 0.0 + (-1)(0) + 0.0$
$= 2{,}5 + 1 - 0{,}5 = 3$ //

$X(3) = Re\ \tilde{X}(0)\cos(2\pi.0.3/4) + Re\ \tilde{X}(1)\cos(2\pi.1.3/4) + Re\ \tilde{X}(2)\cos(2\pi.2.3/4) + Im\ \tilde{X}(0)\sin(2\pi.0.3/4) + Im\ \tilde{X}(1)\sin(2\pi.1.3/4) + Im\ \tilde{X}(2)\sin(2\pi.2.3/4)$
$= (2{,}5)(1) + (-1)(0) + (-0{,}5)(-1) + 0.0 + (-1)(-1) + 0.0$
$= 2{,}5 + 0{,}5 + 1 = 4$ //

$X(n) = \{1, 2, 3, 4\}$